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UGRIC FOREST HUNTERS, THE SIBERIAN KHANATE AND RUSSIAN PEASANT SETTLEMENT IN THE OB-IRTYSH BASIN

Sargat Fortified Towns, the Yugra Principalities and Peasant Colonisation in Western Siberia

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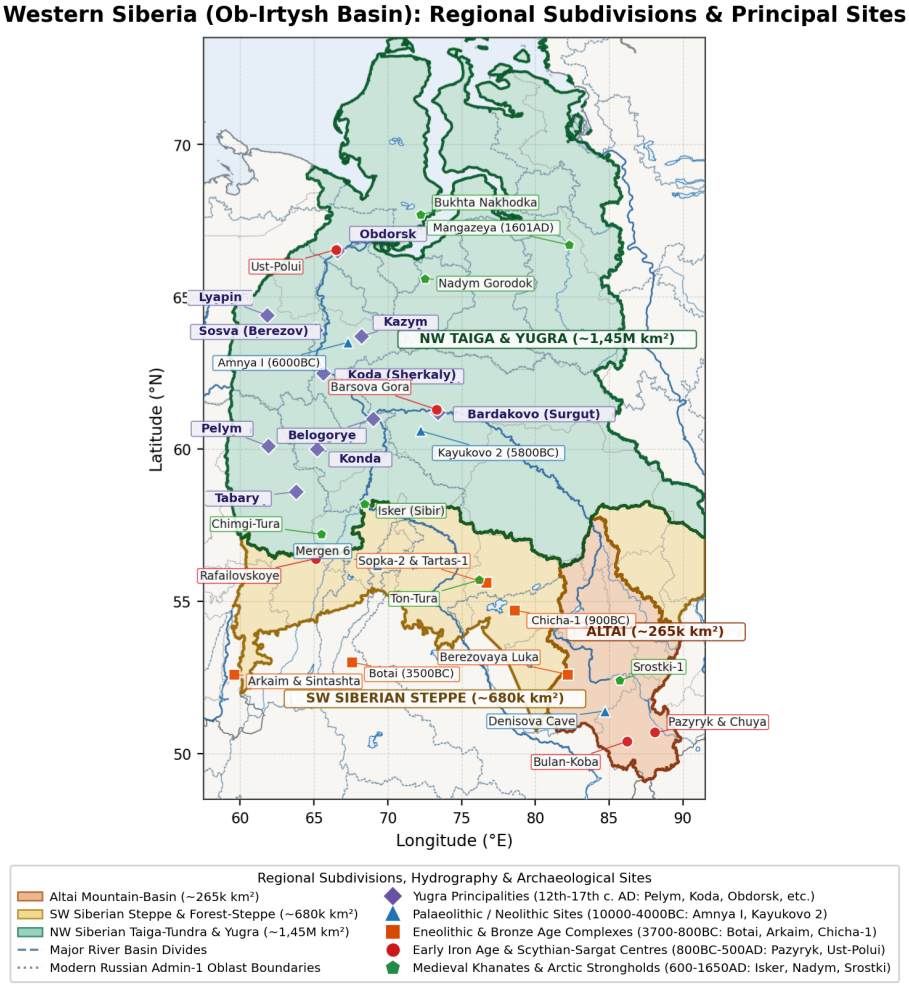


Figure 1. The three regions of the Ob-Irtysh basin (2,4M km^2): the Mountain and Foothill Altai (265k km^2), the South-West Siberian steppe and forest-steppe (680k km^2) and the northern taiga, tundra and Yugra (1,45M km^2).

Abstract

Western Siberia - the 2,4M km^2 drainage basin of the Ob and Irtysh rivers - stretches across three distinct environments: the Altai mountains and foothills (265k km^2), the South-West Siberian steppe and forest-steppe (680k km^2) and the northern taiga, tundra and Yugra (1,45M km^2) [1]. For several millennia

after the Ice Age, fishing and hunting camps along the rivers and lakes supported a small population that grew from ~7,3k in 10000BC to ~20,8k by 6000BC, when the first fortified pit-house villages appeared in the northern forest at Amnya I and Kayukovo 2 [2][3]. The spread of horse-hunting villages at Botai [4], Afanasievo pastoralists in the Altai [5] and Bronze Age copper-smelting towns and farming villages (Arkaim, Kamennyi Ambar and Chicha-1) [6][7] raised the basin's population to ~97,8k in 2000BC and ~189,2k by 800BC. During the Iron Age and early Middle Ages, mounted pastoralists in the Altai (Pazyryk and Srostki cultures) and large fortified towns in the forest-steppe (the Sargat culture around Rafailovskoye) carried the regional total to ~221,2k in 400BC and ~224,8k in 1000AD [8][9][10]. On the eve of Yermak's 1582 campaign, the basin held ~163,5k people, divided between the Siberian Khanate (~39,3k in its core valleys), the ten Ob-Ugrian principalities of Yugra (~21,5k) and the southern steppe and Altai tribes [11][12][13]. Thereafter, Russian peasant settlement along the Tobol-Irtysh river valleys, the Siberian Tract, the Altai mining district and the Trans-Siberian Railway transformed the demographic balance, lifting the population to ~269k in 1700AD, ~855k in 1800AD and ~3,56M by 1900AD [14][15][16][17].

Background

Stretching from the Ural watershed (58°E-60°E) to the Yenisei-Chulym divide (86°E-90°E) and from the mountain valleys of the Russian Altai (49°N) to the Kara Sea coast (73°N), the Ob-Irtysh drainage basin forms the western third of Siberia [1]. As shown in Figure 1, climate and elevation divide Western Siberia into three regions. The first is the Altai region (~265k km^2; 49°N-54°N, 80°E-90°E), covering the high mountain valleys of the Mountain Altai (*Gorny Altai* - the Katun, Biya, Chuya, Argut, Ursul, Chulyshman, Bukhtarma, Ukok and Anui/Denisova valleys) and the northern forest-steppe foothills (*Altaiskoye Priobye* - the Charysh, Alei, Inya and Chumysh valleys). Rich in sheltered winter pastures and copper-tin ores, the Altai supported early herders during the Copper Age (Afanasievo culture), horse-riding nomads during the Scythian (Pazyryk), Hunno-Sarmatian (Bulan-Koba) and Turkic (Srostki) periods and rapid mining and farming growth under the Kolyvano-Voskresensk (Altai) Mining District in the 18th and 19th centuries [5][8][18][19].

The second region is the South-West Siberian Steppe and Forest-Steppe (~680k km^2; 52°N-57,5°N, 61°E-86°E), spanning the Southern Trans-Urals and Tobol-Ishim interfluve (the Tura, Pyshma, Iset, Miass, Tobol, Ubagan and Ishim rivers), the Middle Irtysh (Tara, Om and Osha rivers), the Baraba forest-steppe (Tartas and Kargat rivers, Lakes Chany and Sartlan) and the Kulunda steppe [20][21][22]. This fertile belt of black earth and meadow steppe supported Copper Age horse-hunters (Botai-Tersek), Middle Bronze Age fortified settlements (Sintashta-Petrovka), Late Bronze Age fortified towns (Chicha-1), Early Iron Age chiefdoms (Sargat culture), the Late Medieval Tyumen and Siberian Khanates and the main stream of Russian peasant settlement along the Siberian Tract and Trans-Siberian Railway [4][6][9][13][15][23].

The third region covers the North-West Siberian Taiga, Tundra and Yugra (~1,45M km^2; 57,5°N-73°N, 58°E-86°E), spanning the northern forest, swamps and Arctic tundra of the Lower and Middle Ob (Surgut, Berezov and Obdorsk reaches), the Lower Irtysh (Demyanka and Konda basins), the eastern Ural tributaries (Tavda, Pelym, Lozva, Northern Sosva, Lyapin and Voikar) and the right-bank Arctic rivers (Vakh, Trom-Yugan, Pim, Kazym, Polui, Nadym, Pur, Taz and Yamal) [24][25]. Seasonal runs of whitefish and sturgeon alongside migrating elk and reindeer herds sustained fortified fishing villages in the Neolithic (Amnya I and Kayukovo 2, c. 6000 cal BC), Iron Age metalworking centres (Barsova Gora, Ust-Polui and the Kulai culture) and ten hereditary Ob-Ugrian (Khanty and Mansi) principalities (*knyazhestva*) recorded in 16th- and 17th-century Russian documents [2][3][11][12].

Methodology

Regional population estimates in this paper are built from three main lines of evidence: excavated house counts and floor areas, burial mortality profiles from cemeteries and historical tax and census records.

House Counts and Settlement Area

In the northern forest and tundra, excavations of Early Neolithic (Amnya I, Kayukovo 2), Bronze Age (Barsova Gora) and Iron Age (Beloyarskaya, Kulai, Karym) settlements by V. A. Borzunov [24][26][27], E. N. Dubovtseva [28][29], O. V. Kardash [3] and H. Piezonka [2][30] identify single-chamber pit-houses (*poluzemlyanki* of 15-45 m^2, occupied by families of 5-8 people at ~4,5-5,5 m^2 of heated winter floor space per person) and multi-hearth pit-houses (60-160 m^2, such as at Kayukovo 2 and Barsov Gorodok, occupied by 12-25 people across 2-4 hearth families). In the medieval Ob-Ugrian fortified towns (*gorodishcha* and *gorodki*, 12th-17th c. AD), excavations in permafrost at Nadym Gorodok [31], Ust-Voikary [32], Berezov/Sugmut-Vash [33] and Emder [34] reveal dense rows of surface log cabins (16-36 m^2, housing 5-7 people each) inside timber walls (700-3,5k m^2), surrounded by outer seasonal quarters (*posad*). Assuming an occupancy rate of 0,7 across house depressions and a mean household size of 6,5 persons, settlement population is calculated as:

$$P_{\text{site}} = N_{\text{dwellings}} \times c_{\text{occ}} \times H_{\text{size}}$$

In the steppe and forest-steppe zone, magnetometry surveys and excavations at the Middle Bronze Age Sintashta-Petrovka fortified settlements (2100-1750 cal BC) of Arkaim (60-67 houses), Kamennyi Ambar (46 houses) and Sintashta (35-42 houses) document radial or linear blocks of timber-and-turf row-houses measuring 110-220 m^2 each [6][35]. While early studies suggested 25-30 occupants per house, cemetery and house analyses by A. V. Epimakhov [36][37] and B. Hanks [38] show that roughly one-third of indoor floor area was used for winter stalling of young livestock, wells and copper-smelting furnaces, indicating 10-15 people per house and 450-950 permanent residents per fortified settlement. For the Early Iron Age Sargat settlements (500BC - 400AD), N. P. Matveeva's [9][39] study of Middle Tobol and Iset settlements (Rafailovskoye, Pavlinovo, Bayevo) combines multi-room house floor areas (70-180 m^2) with contemporary burial grounds (705 individuals across 43 cemeteries) to estimate an average household of 10-11 people, with ordinary villages (*selishcha*) housing 250-360 residents and major fortified towns such as Rafailovskoye supporting 1,2k-1,65k inhabitants.

Cemetery Mortality Estimates

In the mountain valleys of the Altai and the open steppes of South-West Siberia, mobile herders' winter camps leave thin cultural layers, whereas burial mounds (*kurgans*) and flat cemeteries are well preserved. Following the standard Acsádi-Nemeskéri life-table formula as applied to Siberian skeletal series by N. P. Matveeva [39], A. V. Epimakhov [36] and N. N. Seregin [18]:

$$P_{\text{living}} = \frac{N_{\text{burials}} \times e_0}{T_{\text{duration}} \times f_{\text{preserved}}}$$

where life expectancy at birth (e_0) is 24,5-27,5 years (derived from skeletal series at Sopka-2, Tartas-1 and 43 Sargat cemeteries where adult life expectancy at age 20 is $e_{20} = 16{,}5$ years and child mortality under age 15 is 38-45%), T_{duration} is the radiocarbon phase duration in years and $f_{\text{preserved}}$ is the estimated burial recovery fraction (0,6-0,8 in large flat cemeteries such as Tartas-1, compared with 0,08-0,15 for large stone burial mounds of the Pazyryk and Sargat elites).

Tax Registers and Census Records

Early Russian fur-tribute registers (*yasachnye knigi*) preserved in the Russian State Archive of Ancient Acts (RGADA, Fund 214, *Sibirsky Prikaz*) record headcounts of adult able-bodied native male tribute payers (*yasachnye lyudi*, aged 18-50). Following B. O. Dolgikh [11], S. V. Bakhrushin [12], E. V. Perevalova [40] and Ya. G. Solodkin [41], total native population is estimated using a multiplier of k = 4,1-4,5 per registered *yasak* payer for the Ob-Ugrian (Khanty and Mansi) principalities (combining Dolgikh's base ratio of 4 with a 5-12% allowance for unregistered dependents and remote forest or tundra camps), a multiplier of k = 4-4,3 applied to registered tribute-paying (*yasachnye*), military-service (*sluzhilye*) and Bukharan merchant (*bukhartsy*) men for the Siberian Tatars across Tobolsk, Tyumen, Turinsk, Tara and Baraba *volosts* [11][13] and a multiplier of k = 4 per tribute payer or armed horseman listed in Russian diplomatic reports (*stateynye spiski*) for the Altai Teleuts, Telengits and Northern Altaian tributaries [11][42].

Between 1719 and 1858, the Russian state conducted ten Poll-Tax Revisions (*revizskie skazki*) counting taxable male "souls" (*revizskie dushi*) among Russian state peasants (*gosudarstvennye krestyane*), Cabinet mining peasants (*pripisnye krestyane*), townspeople and Siberian *yasak* payers (re-registered under Major Ya. Shcherbachev's 1763 *yasak* commission) [14][15][19]. Total population from revision male counts is obtained by doubling male soul totals (incorporating female family members recorded from the 3rd Revision of 1762-1764 onwards) and adding non-taxable groups (Cossacks of the Siberian Host, regular troops, clergy and exiled settlers) [14]. Finally, the First All-Russian Census of 28 January 1897 (*Pervaya vseobshchaya perepis' naseleniya Rossiyskoy Imperii 1897 g.*), analysed at the *uezd* and *volost* level by S. K. Patkanov [16], Z. A. Tychinskikh [17], E. A. Bryukhanova [43] and Yu. V. Semenov [44], provides direct household returns across Tobolsk Governorate, Tomsk Governorate (including the Altai Mining District) and Akmolinsk Oblast.

Results

Altai Region

Early Foragers and Afanasievo Herders

During the Younger Dryas and Early Holocene warming (10000-6000BC), the sheltered lower valleys of the Anui, Katun, Biya and Charysh rivers served as refuges for hunting and fishing groups. Stratified cave and open-air sites at Denisova Cave (Holocene layers 0-8), Kaminnaya Cave, Ust-Karakol-1, Kara-Bom and Tytkesken-2 (along the Middle Katun, where nine cultural layers span the Mesolithic to the Copper Age) show a gradual shift from Final Palaeolithic hunters to Mesolithic and Kuznetsk-Altai Neolithic river fisher-hunters living in pit-houses (16-28 m^2) and using communal cemeteries (Ust-Isha, Solontsy-5, Itkul). Across the Altai region, population grew gradually from ~1,5k (1,2k-1,9k; 0,006 persons/km^2) in 10000BC to ~4k in 6000BC and ~7,5k by 4000BC.

Around 3100-3050 cal BC, the Mountain Altai experienced a major shift with the arrival of the Afanasievo culture (*Afanasievskaya kultura*), established by Western Steppe (Yamnaya-related) herders who introduced domesticated cattle, sheep, goats and horses [5][45]. A series of more than 150 AMS radiocarbon dates compiled by A. V. Poliakov, S. V. Svyatko and N. F. Stepanova [5] shows that Mountain Altai Afanasievo sites cluster between 3100 and 2800 cal BC, slightly earlier than the Minusinsk Basin sites (2900-2500 cal BC). Over 60 burial mound cemeteries (Elo-1/2, Balyktyul, Kurta-2, Nizhnyaya Sooru, Bike-1/3, Kuyum, Aragol, Karasu) containing more than 450 stone-ringed enclosures and 15+ seasonal settlements (Elo-Bashichi, Kara-Tenesh, Malyi Dugan; [45]) indicate ~6k-8k herders in the mountain valleys alongside ~6,5k-8,8k Copper Age Bolshoy Mys inhabitants in the northern foothills, bringing the total Altai population to ~13,5k (10,8k-16,8k; 0,051 persons/km^2) in 3000BC.

Bronze Age Metallurgy and Farming Villages

During the Early and Middle Bronze Age (2200-1700 cal BC), settlement patterns differed between the high mountain valleys and the northern forest-steppe plains [46]. The Karakol culture (2200-1700 cal BC), concentrated in the Ursul and Middle Katun valleys (Karakol, Besh-Ozek, Ozernoye) and known for stone-cist graves decorated with painted stone slabs, supported ~5k-7,5k mountain inhabitants. At the same time, the Elunino culture (2200-1700 cal BC) occupied the Upper Ob forest-steppe and the copper-tin mining foothills of the Rudny Altai [46][47]. At the settlement of Berezovaya Luka on the Alei River (excavated over >2,2k m^2), S. P. Grushin found permanent timber-frame houses (17-20 m^2), bronze-casting workshops and large herds of horses (36%), sheep and goats (37%) and cattle (27%). More than 80 Elunino settlements and cemeteries (Elunino-1, Teleutsky Vzvoz-1, Starodubtsevo) supported ~15k-21k farmers and herders, raising total Altai population to ~23k (18,5k-28,5k) by 2000BC.

Between 1800 and 800 cal BC, Andronovo (Fedorovo) communities (Firsovo-XIV, Kytmanovo, Rubtsovsk) and Late Bronze Age Irmen (*Irmenskaya kultura*) villages (Blizhniye Elbany, Krasnaya Gorka, Kalistratiha, Plotinnaya) built dense chains of river-terrace villages containing 8-15 large multi-hearth pit-houses (80-150 m^2 each, housing 100-200 residents per village). Regional population rose from ~34k in 1500BC to ~48k in 1000BC and ~55k (45k-66k; 0,208 persons/km^2) by 800BC.

Scythian, Hunno-Sarmatian and Turkic Nomads

Following the early Scythian Arzhan-Maiemir phase (850-550BC), nomadic herding expanded across all elevation zones of the Altai during the Scythian-Saka period (550-200BC). Between 1995 and 2014, joint surveys by Ghent University and Gorno-Altaisk State University led by J. Bourgeois and W. Gheyle [8] mapped ~1,25k km^2 of high mountain valleys (Chuya, Kuray, Yustyd, Sebÿstei, Kalanegir, Dzhazator) using satellite imagery and GPS ground checks, recording more than 4,8k surface monuments (3,8-6,5 monuments/km^2 in Yustyd and Sebÿstei), of which 45-55% (>2,2k kurgans) date to the Pazyryk and Kara-Koba phases. Across all habitable valleys of the Russian Altai (an estimated 18k-25k Scythian kurgans built over 350 years, with $e_0 = 26$ years and an elite kurgan representation fraction of 0,08-0,12), the Mountain Altai Pazyryk population numbered ~22k-30k pastoralists, while the northern foothill Bystryanka, Kamen and Staroaleika farming and herding groups supported ~42k-52k inhabitants, giving a combined Altai population of ~68k (56k-82k; 0,257 persons/km^2) in 400BC.

During the Hunno-Sarmatian and Migration Period (200BC - 500AD), studied by N. N. Seregin and S. S. Matrenin [18] from more than 750 excavated graves across 45+ cemeteries (Bulan-Koba IV, Yaloman II, Ust-Edigan, Belyi Bom II, Choburak-I, Dyalyan, Stepushka), the Bulan-Koba culture (200BC - 450AD) and northern foothill Maima communities maintained a regional population of ~64k in 1AD and ~62k in 400AD, marked by frequent weapon burials (~75% of adult men buried with composite bows, iron armour plates and horses).

As the core metalworking region of the First and Second Turkic Khaganates (Kudyrge, Katanda and Kuray phases; ~70k by 800AD), the Altai reached its highest pre-modern population during the Srostki culture (*Srostkinskaya kultura*, 850-1050AD), linked to the eastern wing of the Kimak-Kipchak confederation [10]. Across more than 160 burial mound cemeteries (Srostki-1, Biyke, Filimonovo, Kamen-2, Shadrino) and 40+ foothill villages combining horse herding with millet and barley growing, the regional population reached ~78k (64k-93k; 0,294 persons/km^2) in 1000AD. After 13th- and 14th-century Mongol-era warfare and pasture shifts reduced regional numbers to ~56k by 1400AD, 17th-century Russian tribute records from Kuznetsk (founded 1618) and Tomsk (founded 1604) analysed by B. O. Dolgikh [11] and O. A. Mitko [42] document ~14k-18k Teleuts (“White Kalmyks” of Prince Abak) and Northern Altaian/Shor tributaries (Kumandins, Tubalars, Chelkans, Abintsy) in the Upper Ob and Biya basins alongside ~28k-36k Southern Altai Telengits and Uriankhai (*dvoedantsy* paying dual tribute across 24 Dzungar *zaisanstvos*), giving ~54k

(45k-64k) in 1580AD and ~55k (including early Russian settlers around Kuznetsk) by 1700AD.

The Altai Mining District and Peasant Colonisation

The population of the Altai grew rapidly after the founding of Biysk Fortress (1709), the construction of Akinfiy Demidov's Kolyvan (1727) and Barnaul (1739-1744) copper and silver smelters and Empress Elizaveta Petrovna's 1747 decree transferring the Kolyvano-Voskresensk Mining District (*Kolyvano-Voskresenskiy gorniy okrug*) to the Imperial Cabinet [14][15][19]. Protected by the Kolyvano-Kuznetsk Fortified Line along the Biya, Katun and Charysh foothills, the Imperial Cabinet assigned tens of thousands of state peasants as ascribed mining peasants (*pripisnye krestyane*) who supplied charcoal, timber and grain to the Barnaul, Pavlovsk and Zmeinogorsk silver works, supplemented by exiled Old Believers ("*Polyaki*") settled in the Uba and Bukhtarma valleys [15]. Following the Qing destruction of the Dzungar Khanate in 1755-1756, twelve Southern Altai *zaisans* petitioned for Russian protection under Empress Elizabeth's decree of 2 May 1756 [19]. Regulated by Major Ya. Shcherbachev's 1763 *yasak* commission and later organised into seven Altaian *dyuchinas* and twenty-four Kalmyk and Teleut *volosts* under M. M. Speransky's 1822 Statute on Indigenous Administration (*Ustav ob upravlenii inorodtsev*), the native Altaian, Teleut and Kumandin population stabilised at ~42k-48k [16][19]. Combined with fast-growing Russian villages, total Altai population rose from ~95k (82k-108k) in 1750AD (2nd Revision) to ~185k (165k-205k; ~140k Russians and ~45k native Altaians) by 1800AD (5th Revision, 1795) and ~340k (310k-370k) by 1850AD (9th Revision) [14].

Following the emancipation of Cabinet mining peasants in 1861 and the Imperial decree of 30 July 1865 opening the Altai Cabinet lands to peasant migration from European Russia [23], the fertile black-earth foothills of the Upper Ob became the main farming destination in Siberia. In the First All-Russian Census of 1897 [16][43][44], Biysk *uezd* recorded 337k inhabitants, Zmeinogorsk *uezd* 242,7k and the Upper Ob and foothill *volosts* of Barnaul *uezd* (584,1k across the entire *uezd*, with Barnaul city at 21,1k and Biysk city at 17,2k) contributed ~240k-260k within the Altai boundary, including 49,5k native Turkic inhabitants counted by S. K. Patkanov (42,2k Altaians, Telengits, Kumandins, Tubalars and Chelkans, plus 7,3k Teleuts) [16]. Including peasant arrivals between 1897 and 1900, the population of the Altai region reached ~875k (820k-930k; 3,3 persons/km^2) in 1900AD.

South-West Siberian Steppe and Forest-Steppe

Mesolithic and Neolithic Foragers and Botai Horse-Hunters

Pollen cores and radiocarbon series from the Tobol-Ishim interfluve (Oshukovskoye peatland, Lake Kyrtyma, Lake Mergen) studied by N. E. Ryabogina and V. A. Zakh [20][21] and from the Baraba forest-steppe studied by V. I. Molodin [22][48] show that Boreal and Atlantic warming (8500-5000BC) established birch-pine parklands and shallow fish-rich lakes across the 680k km^2 of the South-West Siberian steppe and forest-steppe. From sparse Mesolithic hunting and fishing camps (Chernoozerye VIa, Kamyshnoye-1, Vengerovo-5, Tartas-3; ~3,8k in 10000BC), population grew during the Early and Late Neolithic (Koshkino, Boborykino, Koksharovsko-Yuryinskaya and Ust-Tartas cultures; Mergen 6 with 20+ pit-houses of 20-50 m^2 and Vengerovo-2A) from ~11k (8,8k-13,8k) in 6000BC to ~21k (16,8k-26k) in 4000BC.

Between 3700 and 3100 cal BC, the southern Ishim-Tobol forest-steppe margin saw the growth of the Botai-Tersek culture, whose economy relied heavily (90-99% of animal bones) on horse hunting, taming and mare's milk (*koumiss*) processing [4][49]. Fluxgate magnetometry surveys led by A. K. Outram and V. F. Zaibert [49] mapped 158-250+ polygonal pit-houses (20-70 m^2) across 15 ha at the Botai type-site (~160 contemporaneous houses holding ~450-800 residents), 54 houses at Krasnyi Yar and 44 houses at Vasilkovka IV, raising total steppe and forest-steppe population to ~35k (28k-43,5k; 0,051 persons/km^2) by 3000BC.

Bronze Age Fortified Towns and Forest-Steppe Villages

During the Early and Middle Bronze Age (2300-1750 cal BC), circular Tashkovo-2 villages in the Lower Tobol and Tura basin (11-13 timber houses of 35-48 m^2 arranged around a central open space, housing 70-100 residents) developed alongside the Odinovo and Krotovo cultures of the Middle Irtysh and Baraba (Sopka-2, Tartas-1 with >800 excavated graves, Vengerovo-2 with 60-110 m^2 houses containing indoor tin-bronze furnaces and the Rostovka and Chernoozerye Seima-Turbino cemeteries; [22][48]). In the Southern Trans-Urals and Upper Tobol-Ui-Miass watershed, Sintashta-Petrovka (2100-1750 cal BC) herders and copper smelters built more than 24 fortified circular and rectangular towns, including Arkaim (2 ha, 60-67 longhouses housing 600-1k residents), Kamennyi Ambar (1,8 ha, 46 longhouses of 150-220 m^2 housing 460-700 residents), Sintashta, Ustye, Stepnoye, Kuysak, Alabuga and Petrovka-2 [6][35][36][37][38]. Combining the 24 fortified towns and surrounding hamlets (~15k-24k inhabitants) with Tashkovo and Odinovo-Krotovo communities across the Irtysh and Baraba plains gives a total steppe and forest-steppe population of ~62k (50k-76k; 0,091 persons/km^2) in 2000BC.

During the Andronovo (Alakul and Fedorovo) period (1750-1400BC) and the subsequent Cherkaskul, Pakhomovo, Suzgun, Barkhatovo and Irmen cultures (1400-900BC), hundreds of unfortified river-terrace villages (5-20 large timber houses of 100-250 m^2 per village) spread across the Tobol, Ishim, Irtysh and Baraba valleys, lifting the steppe and forest-steppe population to ~86k in 1500BC and ~108k in 1000BC. Around 900-700 cal BC (Late Irmen transition), cooler, wetter conditions and contact with northern forest groups (Atlym and Suzgun) led to the building of large fortified towns in Baraba [7]. At Chicha-1 on Lake Malaya Chicha, Russian-German magnetometry and excavations directed by V. I. Molodin, H. Parzinger and J. Schneeweiss [7] uncovered a 5,2 ha fortified town divided by ditches and ramparts into an inner citadel (Zone I) and three outer residential quarters (Zones II-IV) containing 100-110+ contemporaneous pit and timber-frame houses (40-120 m^2 each), supporting ~800-1,25k inhabitants (mean ~1,05k). Regional population in the steppe and forest-steppe zone reached ~118k (96k-142k) in 800BC.

The Sargat Confederacy and the Siberian Khanate

Covering more than 400k km^2 across the Tobol, Iset, Pyshma, Ishim, Middle Irtysh and Baraba forest-steppe, the Sargat culture (*Sargatskaya kultura*, 500BC - 400AD) formed the largest pre-modern population in Western Siberia, widely linked to Ugrian and Iranian mounted pastoralists and forest-steppe farmers [9][39][50]. More than 500 known Sargat sites (~80 fortified *gorodishcha*, ~300 unfortified villages and ~150 burial mound cemeteries) form a three-tier pattern of towns, villages and camps studied by N. P. Matveeva [9][39]. Major fortified towns included Rafailovskoye on the Iset (a 1,8 ha multi-rampart citadel plus a 12 ha outer settlement containing ~155 contemporaneous multi-room houses and supporting 1,2k-1,65k residents; [50]), Pavlinovo, Aktau, Inbereny-IV, Sargatskoye, Bayevo and Abramovo-4; ordinary fortified refuges and villages held 25-35 active households of 10-11 people each (250-360 residents per village); and seasonal herding camps contained 5-10 households (50-100 people). Combining mortality data from 43 Sargat cemeteries (705 individuals; adult life expectancy $e_{20} = 16{,}5$ years, mean adult age at death 36,5 years; [39]) with settlement counts across the Tobol-Irtysh river valleys gives an estimated steppe and forest-steppe population of ~136k (112k-162k; 0,2 persons/km^2) in 400BC and ~138k (114k-165k) in 1AD, declining to ~112k by 400AD during the Migration Period.

Between 600 and 1240AD, Bakalskaya (Tobol-Iset), Yudino (Tura-Tavda) and Ust-Ishim (Middle Irtysh) communities traded and fought with the Kimak-Kipchak Khaganate (whose main seat on the Middle Irtysh is mentioned by medieval Islamic geographers) and Cuman-Kipchak tribes, sustaining ~118k inhabitants in 800AD, ~126k in 1000AD and ~120k in 1200AD. Under the Ulus of Jochi (the Shibanid appanage and the local Taibugid rulers), 14th-century political fragmentation and plague reduced the steppe and forest-steppe population to ~92k by 1400AD. In the 15th and 16th centuries, the region reorganised under the Tyumen Khanate (centred at Chimgi-Tura, modern Tyumen) and the Siberian Khanate

(c. 1495-1598AD) ruled by the Taibugids and the Shibanid Khan Kuchum (r. 1563-1598) [13][51][52].

Excavations of the Khanate's 65+ fortified settlements (*gorodishcha*) reviewed by A. P. Zykov, S. F. Tataurov and D. N. Maslyuzhenko [13][52] show that the capital Isker (Qashliq/Gorod Sibir) occupied a 2-2,6 ha triple-rampart promontory citadel with an outer settlement (~95 structures) housing 450-850 permanent residents (rising to 2k-3,5k during military musters), flanked by regional forts at Chimgi-Tura (500-1k residents), Suzgun, Chuvashsky Mys, Karachin Gorodok, Kullara, Kyzyl-Tura and Ton-Tura (Voznesenskoye on the Om River in Baraba, 300-600 residents). Based on the earliest Russian *yasak* and service-Tatar registers (1590s-1630s) compiled by B. O. Dolgikh [11], the native population of the Siberian Khanate's core territory in c. 1580AD numbered ~34,3k-44,3k (mean ~39,3k: ~9,5k-11k Tobolsk and Vagay Tatars, ~5,8k-6,8k Tyumen and Tura Tatars and Southern Mansi, ~3,2k-4k Tara and Middle Irtysh Tatars, ~3,8k-4,5k Baraba Tatars and ~12k-18k Ishim-Kulunda nomadic subjects), within a total steppe and forest-steppe population of ~88k (72k-105k) including southern Kazakh and Kalmyk steppe camps. Following the founding of Tyumen (1586), Tobolsk (1587), Tara (1594), Turinsk (1600), Yalutorovsk (1659) and Shadrinsk (1662), Russian peasant (*pashennye krestyane*) and Cossack settlement across the Tura, Pyshma, Iset and Tobol valleys added ~145k Russian inhabitants by the 1701 Tobolsk census alongside ~43k Siberian Tatars, Bukharans and southern pastoralists, bringing the steppe and forest-steppe population to ~134k in 1650AD and ~188k (172k-204k; 0,276 persons/km^2) by 1700AD.

Fortified Lines, Peasant Resettlement and the Trans-Siberian Railway

Throughout the 18th and 19th centuries, the South-West Siberian forest-steppe served as the main gateway for Russian peasant settlement in Asia [14][15][23]. Following I. D. Bukhgolts's founding of Omsk Fortress (1716) and the building of the Irtysh and Novo-Ishimskaya fortified lines (Presnogorkovskaya - Petropavlovskaya - Omskaya, 1730s-1752), Russian military and peasant settlement pushed southward from the Tobol-Tura river valleys onto the open black-earth plains of Kurgan, Ishim, Omsk and the Baraba Siberian Tract (*Sibirskiy trakt*) [15]. According to V. M. Kabuzan's [14] and A. D. Kolesnikov's [15] studies of the 1st through 10th Poll-Tax Revisions, the population of the steppe and forest-steppe zone nearly doubled every half-century: rising from ~188k in 1700AD to ~340k (305k-375k) in 1750AD (2nd Revision, 1744-1747), ~640k (585k-695k) in 1800AD (5th Revision, 1795) and ~1,18M (1,09M-1,27M; 1,74 persons/km^2) in 1850AD (9th Revision). Alongside Russian state peasants and Cossacks of the Siberian Line Host, the native and resettled Turkic population (Tobolsk, Tyumen, Tara and Baraba Tatars, Central Asian *Bukhartsy* and Volga-Ural Tatar settlers) expanded from ~43k in 1700AD to ~52k in 1795 and ~64k by the 10th Revision of 1858 [17][53].

After the 1861 peasant emancipation, the 1889 Resettlement Law and the opening of the West Siberian section of the Trans-Siberian Railway (Chelyabinsk - Kurgan - Petropavlovsk - Omsk - Kainsk - Ob, 1892-1896), state-assisted peasant migration into the Tobol, Ishim and Baraba steppes accelerated sharply [23][54]. In the 1897 Census [16][17][43][44], the eight southern forest-steppe *uezds* of Tobolsk Governorate recorded 1,4M inhabitants (Ishim *uezd* 269k, Kurgan 260,1k, Tyukalinsk 208,7k, Yalutorovsk 188,6k, Tara 159,7k, Tobolsk 127,9k, Tyumen 121,4k and Turinsk 68,6k, including 57k Tatars and Bukharans in Tobolsk Governorate [17][53]), while Kainsk *uezd* (187,1k) and the northern Baraba-Kulunda *volosts* of Tomsk Governorate added ~512k, and the northern forest-steppe districts of Akmolinsk Oblast (Petropavlovsk *uezd* 227,5k, Omsk *uezd* 100,5k with Omsk city at 37,4k and adjoining Ishim-Irtysh *volosts*) accounted for ~580k-620k residents. Including railway arrivals between 1897 and 1900, the total population of the South-West Siberian Steppe and Forest-Steppe reached ~2,65M (2,5M-2,8M; 3,9 persons/km^2) in 1900AD.

North-West Siberian Taiga, Tundra and Yugra

Early Neolithic Fortified Settlements: Amnya I and Kayukovo 2

As post-glacial forests spread across the Lower Ob and Konda-Sosva basins, Mesolithic hunter-fishers occupied pine-covered river terraces (Lugovskoye, Sumpanya-IV/VI, Korstovo, Komudvany; ~2k across 1,45M km^2 in 10000BC, rising to ~3,2k in 8000BC). Around 6100-5500 cal BC, Early Neolithic communities in the West Siberian taiga began building fortified promontory villages (260-1,9k m^2) enclosed by timber palisades, earthen banks and ditches, as studied by V. A. Borzunov [24][27], E. N. Dubovtseva [28][29], O. V. Kardash [3] and H. Piezonka [2][30]. At Amnya I and Amnya II in the Kazym River basin (63,5°N), AMS radiocarbon dating published by H. Piezonka, N. M. Chairkina, E. N. Dubovtseva and L. L. Kosinskaya [2][28] dates the inner defensive ditch, rampart, palisade and House 1 to 6100-5900 cal BC, making Amnya I the oldest dated fortified settlement in northern Eurasia; the site consists of 10 semi-subterranean pit-houses (15-40 m^2, 0,6-1,8 m deep) behind three defensive ditch-and-bank lines (Amnya I), together with 10 unfortified outer pit-houses (Amnya II) 50 m to the east, housing ~40-75 residents (mean ~60) who controlled seasonal fish-weir (*zapor*) and elk-crossing sites.

At Kayukovo 2 on the Bolshoy Salym - Bolshoy Yugan watershed, excavated by O. V. Kardash, N. M. Chairkina, E. N. Dubovtseva and H. Piezonka [3] and AMS-dated to 6000-5500 cal BC, a circular fortified settlement (35 x 35 m, ~1,15k m^2 inner enclosure within a 0,25 ha site) was surrounded by a timber wall and ring ditch and contained 5 outer pit-houses (16-32 m^2) linked by sunken passages to a central communal building (House 6), housing ~30-55 residents (mean ~42). Together with Imnegan, Sumpanya IV/VI and Et-to I/II (on the Nadym-Pur watershed at 63,2°N), more than 300 Neolithic pit-house sites across the northern taiga and tundra supported ~5,8k (4,6k-7,2k) inhabitants in 6000BC and ~8,8k (7k-11k) by 4000BC.

Copper Age, Bronze Age and Iron Age Centres

During the Copper and Bronze Ages (4000-800BC), comb-pit pottery groups (Sartyn'ya, Shapkul, Chuovsky) living in multi-hearth winter pit-houses (40-90 m^2) supported ~10,8k inhabitants in 3000BC. During the Seima-Turbino period (2150-1900 cal BC; Satyga XVI sanctuary) and the Late Bronze Age Atlym culture (1200-800BC), large semi-subterranean houses (50-120 m^2) spread north to the Arctic Circle, raising the northern taiga and tundra population from ~12,8k in 2000BC to ~15,5k in 1000BC. Along an 8 km bluff on the right bank of the Middle Ob near Surgut, the Barsova Gora complex studied by Yu. P. Chemyakin and A. P. Zykov [55][56] preserves 66 fortified settlements (*gorodishcha*), ~300 unfortified villages (*selishcha*), 8 ritual sites and ~3k house depressions spanning 7k years from the Neolithic to the 17th century AD.

Between 500BC and 450AD, the Kulai culture (*Kulaiskaya kultura*) spread across the Middle and Lower Ob (Sarovskoye, Barsov Gorodok I/32; [56]). At Ust-Polui (260BC - 140AD, on the Polui River cape in modern Salekhard at 66,5°N), excavations by N. V. Fedorova, A. V. Gusev, R. J. Losey and E. V. Vodyasov [57][58][59][60] uncovered a ditch-and-palisade sanctuary and metalworking site (~4,2k m^2) containing the earliest evidence of iron smelting and forging in the Arctic [57][58], carved bone harness toggles for reindeer sleds [60] and the remains of more than 115 domesticated sledging and sacrificial dogs [59]. During the subsequent Karym phase (350-550AD) studied by V. A. Borzunov and Yu. P. Chemyakin [26], fortified taiga settlements multiplied along the Surgut and Lower Ob rivers, with regional population growing from ~17,2k in 400BC to ~18k in 1AD and ~18,6k in 400AD.

Medieval Yugra and the Ten Ob-Ugrian Principalities

Between the 8th and 16th centuries AD (Lower Ob culture: Kintusovskiy, Oronturskiy, Emder and Saygatinskiy stages; [34][55]), trade in northern furs with Volga Bulgaria, Great Novgorod (recorded in Gyuryata Rogovich's 1096AD account and the Novgorodian expeditions of 1187 and 1193) and the Golden

Horde supported the rise of ten hereditary Ob-Ugrian (Mansi/*Vogul* and Khanty/*Ostyak*) principalities (*knyazhestva*) [11][12][40][41][61]. Analysis of 25,9k fish bones from 17 Late Holocene sites (Nadym Gorodok, Ust-Voikary, Sherkaly-1, Bukhta Nakhodka, Berezov) by T. V. Lobanova and D. O. Gimranov [25] confirms that large-scale seasonal catching and drying (*yukola*) of pike, ide, sturgeon, nelma and muksun supplied the food stores needed for permanent fortified timber towns (*gorodki*). Among the best-excavated princely centres, Sherkaly-1 (*Shorkarskiy Gorodok* in the Koda Principality) comprised three fortified hill enclosures (~1,2 ha) with 40+ surface log cabins fitted with stone ovens (*pechi-kamenki*) and smithies (~200-350 residents; [55]); Emder (Endyrskoye-1) preserved nine successive building layers (11th-16th c. AD, ~2,4k m² citadel and outer settlement, ~120-220 residents; [34]); Ust-Voikary (*Voikarsky Gorodok*) and Berezov (*Sugmut-Vash*) contained permafrost-preserved log towns with ~30 residential and storage buildings (~100-165 residents; [32][33]); and in the far north, Nadym Gorodok (1200-1730AD, a 700 m² timber fort of the *Bolshaya Karacheya* clan with 6 log apartment blocks) housed 40-75 winter residents [31], alongside Bukhta Nakhodka (a 1213-1332AD tree-ring dated Arctic *Sikhirtya* coastal fort on the Yamal Peninsula with 25-45 residents).

No.	Principality	Ethnic Group	Ruling Dynasty	Main Fortified Towns (N)	Adult Male Yasak Payers	Armed Warriors	Estimated Pop. (k = 4,1-4,5)
1	Pelym (Pelymskoe)	Western Mansi (Voguls)	Asyka, Yumshan, Ablegirim	Pelym, Ust-Lozva, Vagil (7)	365-410	400-500	1,8k-2,2k
2	Tabary & Koshuki	Southern Plough Mansi	Prince Bochay (Tabary)	Tabary, Koshuki, Chubarov (4)	177-215	150-200	750-950
3	Kondia (Kondin-skoe)	Konda Mansi & Khanty	Prince Agai / Kondinsky	Kartauzh, Levushi, Satyga (7)	235-340	250-350	1,1k-1,5k
4	Kod (Koda & Emder)	Middle-Lower Ob Khanty	House of Alachev (Moldan, Igichey)	Koda, Shorkar, Vezhakar, Emder (14)	620-730	350-450	3k-4,5k
5	Belogorye (Ob-Irtysh)	Confluence Khanty	Prince Samar (d. 1582)	Samarovo, Belogorye, Tsingaly (7)	340-410	250-320	1,5k-1,85k
6	Kazym (Kazym-skoe)	Northern Khanty & Nenets	Kazym / Moldanov Princes	Yuilsk, Polnovat, Amnya (4)	185-220	120-160	800-980
7	Sosva (Sugmut-Vash)	Northern Mansi & Khanty	Sosva Princes (Berezov)	Sugmut-Vash, Iskar, Sartyn'ya (6)	320-370	200-260	1,35k-1,6k
8	Lyapin & Kunovat	Sygva Mansi & Khanty	Luguyev Dynasty (Luguy, Shatrov)	Lombovozh, Lopmus, Ust-Voikary (6)	430-490	250-320	1,8k-2,15k
9	Obdorsk (Pulnovat-Vash)	Arctic Khanty & Nenets	House of Taishin (Vasily, Mamruk)	Pulnovat-Vash, Sobsky, Muzhi (5)	380-450 (760 w/ Nenets)	300-400	1,77k (3,2k w/ Nenets)

10	Bardakovo & Surgut Ob	Eastern (Surgut) Khanty	Prince Bardak & Prince Tonya	Bardakov, Monkys-Uriy, Pim (10)	390 Core (850 All Surgut)	250-350	1,65k (3,55k All Surgut)
-	Unattached Tura/ Nadym/Taz	Tura Mansi & Taz Nenets	Bolshaya Karacheya & Tura Murzy	Nadym Gorodok, Tura Yurts (5)	550-680	200-300	2,3k-2,9k
Total	All Yugra & NW Taiga	Khanty, Mansi & Nenets	10 Hereditary Princely Houses	65-75 Active Fortified Gorodki	4,68k-5,41k	2,72k-3,61k	19,35k-23,88k (Mean 21,5k)

Table 1. Ruling dynasties, fortified timber towns (gorodki), registered adult male fur-tribute (yasak) payers, armed warriors and total native populations across the ten Ob-Ugrian principalities and neighbouring tundra districts on the eve of Russian rule.

As shown in Table 1 and Figure 2, summing across all ten principalities and neighbouring Tura Mansi and Nadym/Taz Nenets districts gives 16,1k-19,8k Ob-Ugrians (~9,8k Khanty and ~6,4k Mansi) plus ~2,8k-3,8k North-West Siberian Tundra and Forest Nenets (*Samoyeds*) in 1580-1625AD, matching B. O. Dolgikh's [11] estimate of ~21,5k (19,35k-23,88k) native inhabitants across Yugra and the north-western taiga (rising to ~26k by 1700AD when ~4,5k Russian Cossacks and townsmen in Berezov, Surgut, Pelym, Obdorsk and Mangazeya are included).

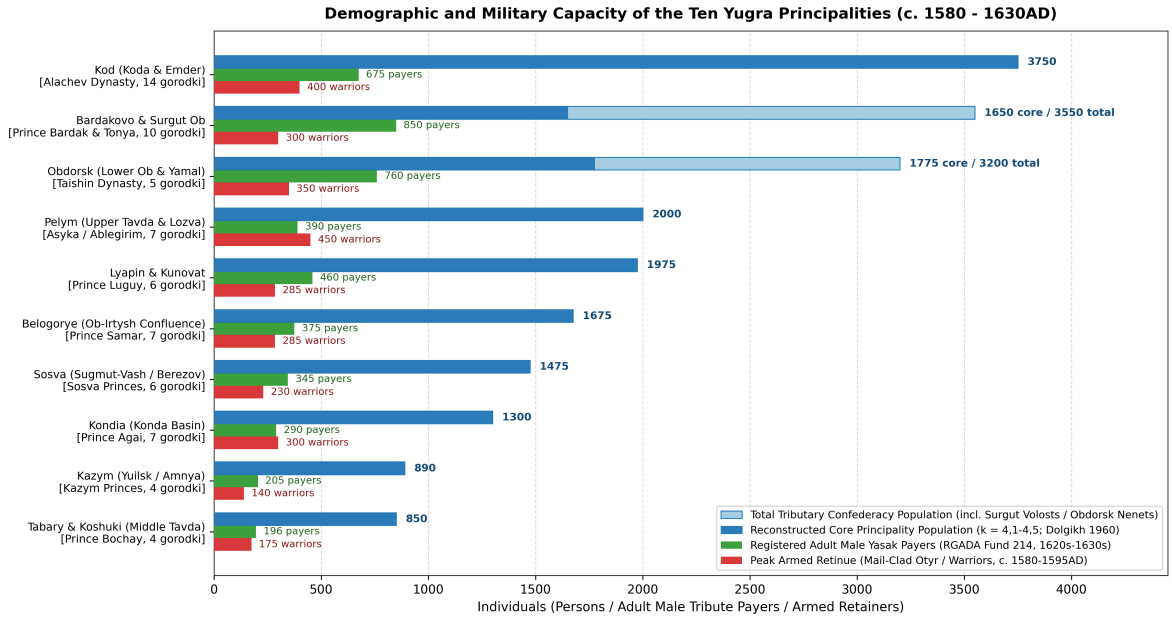


Figure 2. Panel A compares the total population of each Ob-Ugrian principality and unattached tundra district (calculated at k = 4,1-4,5 persons per registered tribute payer, totalling 21,5k native inhabitants). Panel B contrasts adult male yasak payers from the early Sibirsky Prikaz rolls against armed warriors reported in 16th- and early 17th-century Russian campaign accounts.

The Berezov and Surgut Districts under Imperial Rule

During the 18th and 19th centuries, the northern forest and tundra basins of the Lower and Middle Ob were administered mainly through Berezov and Surgut *okrugs* (*uezds*) of Tobolsk Governorate, alongside the northern *volosts* of Turinsk (Pelym) and Verkhoturye [14][16][62]. Following Peter I's decrees of 1706-1714 and the missionary journeys of Metropolitan Filofey (Leshchinsky) between 1712 and 1727 [62], most Khanty and Mansi river communities were baptised, and the old fortified princely *gorodki* (such as Nadym Gorodok, abandoned c. 1730; [31]) gave way to open winter log-cabin villages (*zimovya*) and seasonal birch-bark fishing camps (*yurts*). In Obdorsk, however, the hereditary Taishin princes kept charge of Lower Ob Khanty and Yamal Nenets *yasak* collection until 1886 [40][61]. Across the 2nd (1744-1747), 5th (1795) and 9th (1850) Revisions [14][62], slow growth among northern Khanty and Tundra Nenets reindeer herders balanced the gradual assimilation of southern Mansi and Konda villages into surrounding Russian peasant parishes, lifting total northern taiga and tundra population from ~26k in 1700AD to ~28k in 1750AD, ~30k in 1800AD and ~32k in 1850AD.

In the First All-Russian Census of 1897, studied in detail at the *volost* and clan level by S. K. Patkanov [16], the full *okrug* tally for Berezov *Okrug* (including nomadic Yamal and Kazym tundra *volosts* beyond the 11,4k settled residents) recorded 21,4k inhabitants (72,9% native Khanty, Mansi and Nenets, alongside Russian townsmen in Berezov [1,1k] and Obdorsk and migrant Izhma Komi-Zyrian reindeer herders), while Surgut *Okrug* recorded 7,7k inhabitants (72,1% native Khanty and Forest Nenets, with 1,1k residents in Surgut town). Including the Ob-Ugrian and Selkup *volosts* of northern Tobolsk (Samarovo), Turinsk (Pelym and Konda) and the Upper Taz [16][63], the 1897 Census counted 19,7k Khanty (*Ostyaks*), 7,7k Mansi (*Voguls*) and ~4,5k West Siberian Tundra and Forest Nenets (*Samoyeds*) (~31,8k native inhabitants across the northern taiga and tundra), together with ~3,2k Russians and Komi-Zyrians, giving a total population of ~35k (32k-38k; 0,024 persons/km^2) in 1900AD.

Population Estimates

Excavations and magnetometry surveys at 14 benchmark settlements across Western Siberia (Figure 3 and Table 3) show a clear contrast in settlement scale between the northern forest and the southern forest-steppe. In the northern taiga and Yugra, Early Neolithic pit-house forts (Amnya I and Kayukovo 2) and medieval Ob-Ugrian log towns (Sherkaly-1, Emder, Ust-Voikary and Nadym Gorodok) enclosed 0,12-1,2 ha and housed 40-350 winter residents. In the southern steppe and forest-steppe, Botai horse-hunting villages (450-800 residents), Sintashta-Arkaim copper-smelting forts (460-1k residents), Late Bronze Age Chicha-1 (800-1,25k residents across 100-110 houses) and Early Iron Age Sargat centres such as Rafailovskoye (1,2k-1,65k residents across 13,8 ha) reached five to ten times the size of northern forest forts.

Combining these excavated settlement benchmarks with burial mortality tables, 17th-century *yasak* registers, 18th- and 19th-century Poll-Tax Revisions and the 1897 Census across the three regions of Western Siberia gives the population and density series from 10000BC to 1900AD shown in Figure 4 and Table 2.

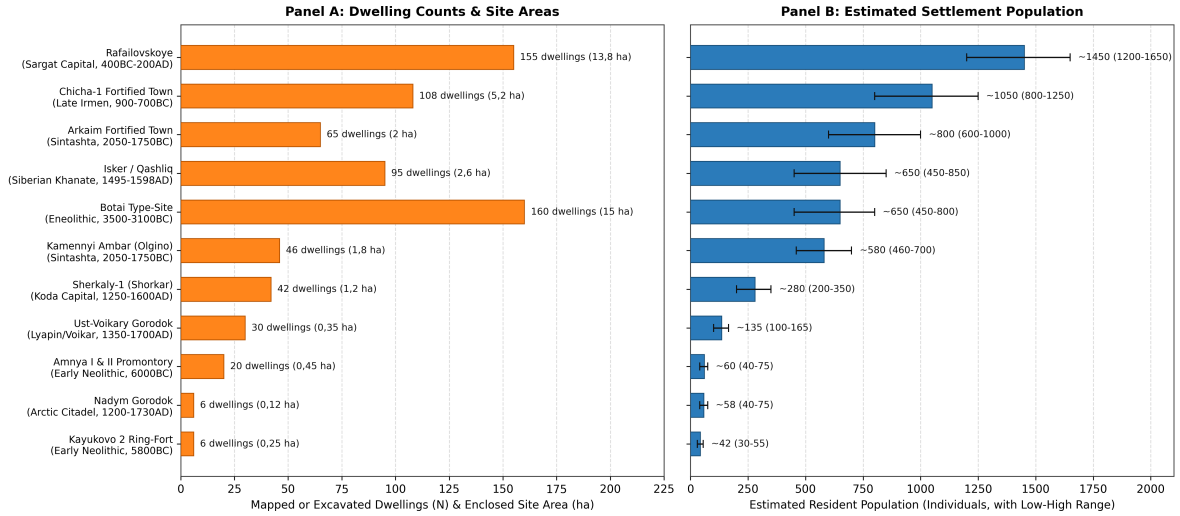


Figure 3. Panel A compares the enclosed areas (in hectares) and excavated house counts of 14 benchmark settlements from Early Neolithic Amnya I and Kayukovo 2 to the Siberian Khanate capital of Isker. Panel B shows the resulting settlement population estimates, peaking at 800-1,25k inhabitants in Late Bronze Age Chicha-1 and 1,2k-1,65k in Early Iron Age Rafailovskoye.

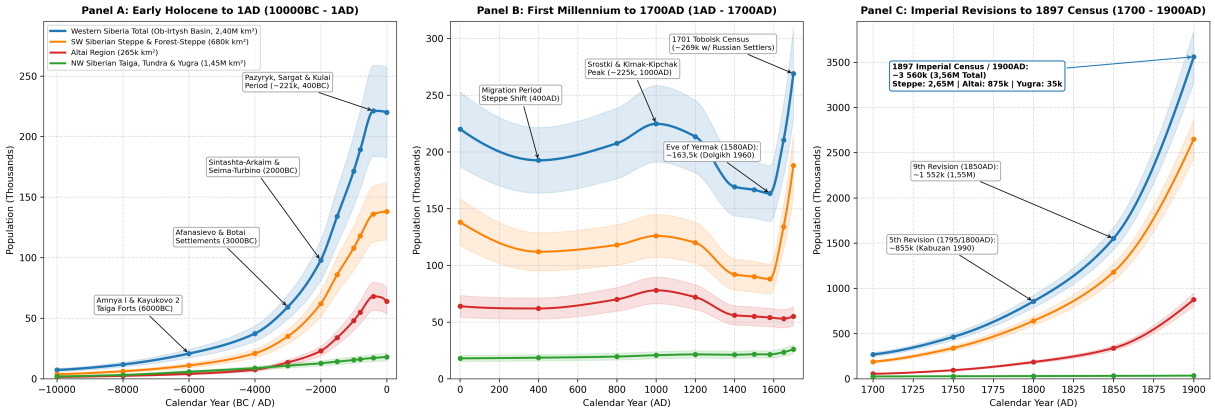


Figure 4. Panel A traces the pre-modern population of Western Siberia from Early Neolithic fishing camps through the Bronze Age (189,2k in 800BC), Sargat-Pazyryk Iron Age (221,2k in 400BC) and Srostki-Kimak medieval (224,8k in 1000AD) peaks to the eve of Yermak's campaign (163,5k in 1580AD). Panel B shows the rapid expansion under Russian rule as peasant settlement along the Siberian Tract, the Altai mining district and the Trans-Siberian Railway raised the regional total from 269k in 1700AD to 3,56M by 1900AD.

Date	Altai Pop. (265k km²)	Altai Dens.	Steppe Pop. (680k km²)	Steppe Dens.	Taiga/ Yugra Pop. (1,45M km²)	Taiga Dens.	Western Siberia Total (2,4M km²)	Main Cultural or Historical Period
10000BC	1,5k (1,2k-1,9k)	0,006	3,8k (3k-4,8k)	0,006	2k (1,5k-2,6k)	0,001	7,3k (5,7k-9,3k)	Final Palaeolithic / Early Mesolithic (Denisova, Chernooz-erye)
8000BC	2,4k (1,9k-3k)	0,009	6,2k (4,9k-7,8k)	0,009	3,2k (2,5k-4,1k)	0,002	11,8k (9,3k-14,9k)	Boreal Mesolithic Hunter-Fishers (Kaminnaya, Sumpanya, Vengerovo-5)
6000BC	4k (3,2k-5k)	0,015	11k (8,8k-13,8k)	0,016	5,8k (4,6k-7,2k)	0,004	20,8k (16,6k-26k)	Early Neolithic Fortified Sites (Amnya I, Kayukovo 2, Koshkino)
4000BC	7,5k (6k-9,4k)	0,028	21k (16,8k-26k)	0,031	8,8k (7k-11k)	0,006	37,3k (29,8k-46,4k)	Late Neolithic / Early Eneolithic (Bobo-rykino, Ust-Tartas)
3000BC	13,5k (10,8k-16,8k)	0,051	35k (28k-43,5k)	0,051	10,8k (8,6k-13,5k)	0,007	59,3k (47,4k-73,8k)	Afanasievo Pastoralists (Altai) & Botai Horse Villages (Steppe)
2000BC	23k (18,5k-28,5k)	0,087	62k (50k-76k)	0,091	12,8k (10,2k-15,8k)	0,009	97,8k (78,7k-120,3k)	Sintashta-Arkaim Forts, Elunino, Krotovo & Seima-Turbino
1500BC	34k (27,5k-41,5k)	0,128	86k (70k-105k)	0,126	14,2k (11,5k-17,5k)	0,01	134,2k (109k-164k)	Andronovo (Alakul & Fedorovo) Farming & Herding Expansion

1000BC	48k (39k-58k)	0,181	108k (88k-130k)	0,159	15,5k (12,6k-19k)	0,011	171,5k (139,6k-207k)	Late Bronze Irmen, Suzgun, Barkhatovo, Atlym & Barsova Gora
800BC	55k (45k-66k)	0,208	118k (96k-142k)	0,174	16,2k (13,2k-19,8k)	0,011	189,2k (154,2k-227,8k)	Chicha-1 Fortified Town (108 Houses) & Early Scythian Period
400BC	68k (56k-82k)	0,257	136k (112k-162k)	0,2	17,2k (14k-21k)	0,012	221,2k (182k-265k)	Pazyryk Culture (Altai), Sargat Confederacy (Steppe), Kulai (Taiga)
1AD	64k (52k-77k)	0,242	138k (114k-165k)	0,203	18k (14,8k-21,8k)	0,012	220k (180,8k-263,8k)	Bulan-Koba (Altai), Late Sargat (Rafailovskoye), Ust-Polui (Arctic)
400AD	62k (50k-75k)	0,234	112k (92k-134k)	0,165	18,6k (15,2k-22,5k)	0,013	192,6k (157,2k-231,5k)	Migration Period (Choburak-I / Dyalyan, Potchevash, Karym)
800AD	70k (58k-84k)	0,264	118k (96k-142k)	0,174	19,6k (16k-23,5k)	0,014	207,6k (170k-249,5k)	First & Second Turkic Khaganates, Early Kimak & Kintusovskiy
1000AD	78k (64k-93k)	0,294	126k (104k-150k)	0,185	20,8k (17k-25k)	0,014	224,8k (185k-268k)	Srostki Culture & Kimak-Kipchak Peak; Early Yugra Gorodki

1200AD	72k (59k-86k)	0,272	120k (98k-144k)	0,176	21,5k (17,8k-25,6k)	0,015	213,5k (174,8k-255,6k)	Cuman-Kipchak, Bakalskaya, Ust-Ishim, Sherkaly-1 & Emder
1400AD	56k (46k-67k)	0,211	92k (75k-110k)	0,135	21,2k (17,5k-25,2k)	0,015	169,2k (138,5k-202,2k)	Ulus of Jochi/Shiban Steppe Contraction; Bukhta Nakhodka
1500AD	55k (45k-66k)	0,208	90k (74k-108k)	0,132	21,6k (18k-25,5k)	0,015	166,6k (137k-199,5k)	Tyumen & Siberian Khanates; Four Oirats; 10 Yugra Polities
1580AD	54k (45k-64k)	0,204	88k (72k-105k)	0,129	21,5k (19,3k-23,9k)	0,015	163,5k (136,3k-192,9k)	Eve of Yermak (Isker, 10 Yugra Principalities, Teleut Uluses)
1650AD	53k (44k-62k)	0,2	134k (115k-154k)	0,197	23,5k (20,5k-26,8k)	0,016	210,5k (179,5k-242,8k)	Russian Agricultural Settlement (Tobolsk, Tyumen, Tara, Surgut)
1700AD	55k (48k-62k)	0,208	188k (172k-204k)	0,276	26k (22,5k-29,5k)	0,018	269k (242,5k-295,5k)	1701 Tobolsk Census (151k Russians + 118k Indigenous)
1750AD	95k (82k-108k)	0,358	340k (305k-375k)	0,5	28k (24,5k-31,5k)	0,019	463k (411,5k-514,5k)	2nd Revision (1744-1747); Irtysh/Ishim Lines & Demidov Works
1800AD	185k (165k-205k)	0,698	640k (585k-695k)	0,941	30k (26,5k-33,5k)	0,021	855k (776,5k-933,5k)	5th Revision (1795); Kolyvano-Voskresensk District & Siberian Tract

1850AD	340k (310k-370k)	1,28	1,18M (1,09M-1,27M)	1,74	32k (28,5k-35,5k)	0,022	1,55M (1,43M-1,68M)	9th Revision (1850); State Peasant Resettle-ment into Tobolsk/ Tomsk
1900AD	875k (820k-930k)	3,3	2,65M (2,5M-2,8M)	3,9	35k (32k-38k)	0,024	3,56M (3,35M-3,77M)	1897 First All-Russian Census & Trans-Siberian Railway Boom

Table 2. Population estimates, uncertainty ranges and densities for the Altai, the South-West Siberian steppe and forest-steppe and the northern taiga and Yugra at 24 benchmark dates, with combined totals across the 2,4M km^2 Ob-Irtysh basin.

Limitations

Three main sources of uncertainty affect the population ranges reported in Table 2. First, because northern forest hunter-fisher groups periodically moved their winter pit-house villages (*poluzemlyanki*) and summer birch-bark fishing camps (*yurts*) along shifting river channels, surface counts of house depressions (~3k at Barsova Gora) build up over several centuries [24][30]; applying a 70% occupancy factor within single-phase fortified enclosures avoids overstating winter village populations. Second, skeletal studies of 43 Sargat burial grounds (705 individuals; [39]) and Mountain Altai Pazyryk and Bulan-Koba cemeteries [8][18] show that infant and child burials (under age 15) are under-represented in large burial mounds compared with flat communal cemeteries (Sopka-2, Tartas-1; [48]), requiring life-table adjustments (e_0 of 25-27 years and a burial recovery fraction of 0,08-0,15 for elite kurgan fields). Third, fish-rich diets across the Tobol-Ishim lakes, Baraba and the Lower Ob can make radiocarbon dates on human bone collagen and pottery food crusts appear 200-500 years too old [2][5][21], so all chronological benchmarks in this paper rely on wood, charcoal or land-animal bone dates; in the early Imperial period, the 1st and 2nd Poll-Tax Revisions (1719-1747) also under-counted runaway peasants, Old Believers in the Altai foothills and remote tundra camps by an estimated 8-15% before the 5th Revision (1795) and 1897 Census [14][16].

Conclusion

Across the 2,4M km^2 Ob-Irtysh basin, population grew through five distinct stages between the end of the Ice Age and the close of the 19th century. Early Neolithic fishing and hunting communities in the northern forest and forest-steppe (Amnya I and Kayukovo 2) raised regional numbers to ~20,8k by 6000BC, before Copper and Bronze Age herding, farming and copper smelting (Afanasievo, Botai, Sintashta-Arkaim, Andronovo, Irmen and Chicha-1) lifted the total from ~59,3k in 3000BC to ~189,2k in 800BC. During the Iron Age and early Middle Ages, mounted pastoralists and fortified forest-steppe towns (Pazyryk, Sargat, Kulai, Bulan-Koba and Srostki/Kimak-Kipchak) sustained ~220k-224,8k inhabitants

between 400BC and 1000AD. After late medieval warfare and plague reduced the basin to ~163,5k on the eve of Yermak's 1582 campaign - including ~39,3k in the core of the Siberian Khanate and ~21,5k across the ten Ob-Ugrian principalities of Yugra - Russian peasant and mining settlement transformed Western Siberia, expanding the population from ~269k in 1700AD to ~855k in 1800AD and ~3,56M by 1900AD.

Appendix

Settlement Estimates

Settlement	Region	Calibrated Date	Area	Excavated / Mapped Dwellings	Estimated Pop. (Range)	Primary Reference
Amnya I & II (Kazym Basin)	Taiga / Yugra	6100-5900 cal BC	0,45 ha	20 pit-houses (15-40 m^2)	40-75 (Mean 60)	Piezonka et al. 2023; Dubovtseva 2020
Kayukovo 2 (Salym-Yugan)	Taiga / Yugra	6000-5500 cal BC	0,25 ha	6 ring-houses (16-32 m^2)	30-55 (Mean 42)	Kardash et al. 2020; Borzunov 2012
Botai Type-Site (Ishim Steppe)	Steppe	3700-3100 cal BC	15 ha	158-250+ pits (160 active)	450-800 (Mean 650)	Outram et al. 2009, 2021
Arkaim (Trans-Ural Steppe)	Steppe	2050-1750 cal BC	2 ha	60-67 radial longhouses	600-1k (Mean 800)	Koryakova & Epimakhov 2007
Kamennyi Ambar (Olgino)	Steppe	2050-1750 cal BC	1,8 ha	46 row longhouses (150-220 m^2)	460-700 (Mean 580)	Krause et al. 2019; Epimakhov 2009
Berezovaya Luka (Alei / Ob)	Altai	2100-1800 cal BC	2,2 ha	18-25 frame-post houses	120-180 (Mean 150)	Grushin 2011; Dashkovskiy et al. 2024
Chicha-1 Fortified Town (Baraba)	Steppe	900-700 cal BC	5,2 ha	100-110+ houses (Zones I-IV)	800-1,25k (Mean 1,05k)	Molodin, Parzinger & Schneeweiss 2009
Rafailovskoye (Iset Valley)	Steppe	400BC - 200AD	13,8 ha	155 multi-room houses	1,2k-1,65k (Mean 1,45k)	Matveeva 1999, 2000; Chikunova 2005
Ust-Polui (Salekhard, Arctic)	Taiga / Yugra	260BC - 140AD	0,42 ha	Sanctuary + smithy enclosure	Regional catchment 1,5k-2,5k	Losey et al. 2018, 2020; Vodyasov 2017
Sherkaly-1 / Shorkar (Koda)	Taiga / Yugra	1250-1600AD	1,2 ha	42 log cabins w/ stone ovens	200-350 (Mean 280)	Zykov 2012; Solodkin 2015
Emder (Endyrskoye-1)	Taiga / Yugra	1100-1600AD	0,35 ha	25-35 citadel + posad cabins	120-220 (Mean 170)	Zykov & Koksharov 2001
Ust-Voikary (Voikar Gorodok)	Taiga / Yugra	1350-1720AD	0,35 ha	30 permafrost log cabins	100-165 (Mean 135)	Vizgalov & Parkhimovich 2008

Nadym Gorodok (Lower Nadym)	Taiga / Yugra	1200-1730AD	0,12 ha	6 apartment log-blocks	40-75 (Mean 58)	Kardash 2009; Lobanova 2016
Isker / Qashliq (Gorod Sibir)	Steppe	1495-1598AD	2,6 ha	Citadel + posad (95 structs)	450-850 (2k-3,5k assembly)	Tataurov & Maslyuzhenko 2016, 2018

Table 3. Calibrated radiocarbon or historical dates, enclosed areas, mapped dwellings and estimated resident populations for 14 excavated reference settlements across Western Siberia.

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